

ANTHONY HICKS III

Senior Software Engineer

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EXPERIENCE

Senior Lead Software Engineer

05/2024 – Present

JPMorgan Chase & Co.

- Optimized an Apache Spark batch job that processes 850 million rows, reducing the processing time by 75% from 6 hours to 1.5 hours.
- Architected the system to reduce the time to enable an advertising campaign from 2 weeks to several hours.
- Led development of effort to serve personalized advertisement campaigns to millions of customers.
- Reduced time-to-market for advertising campaigns by leading the design of a domain-specific language (DSL) that enables external systems to automate advertisement targeting across several lines of business within the firm.
- Set the technical direction for the advertising platform modernization effort across 3 teams (~20 engineers) globally, guiding execution and aligning cross-team dependencies.
- Standardized technical documentation practices across teams by introducing a system-wide documentation framework based on ADRs and RFCs, reducing miscommunication across teams and maintaining a history of architectural decisions.

Software Engineer

07/2022 – 03/2024

Google

- Created a system to prevent breaking changes from entering Google Search's critical features, using Python, Java, gRPC, Google PubSub, and Google Protobufs.
- Designed and implemented an alerting system to notify teams of potential breaking changes entering Google Search.
- Built stakeholder dashboards to track the number of breaking changes that were prevented.
- Led the creation of the team's onboarding process and onboarded the first external user to the infrastructure.
- Automated the process of catching breaking changes that could affect Google Search's Generative AI (AI Overview Section).

Vice President of Software Engineering

02/2021 – 07/2022

JPMorgan Chase & Co.

- Maintained an SLO of the Spring Boot RESTful API at 250 milliseconds even as we continued to develop new features and scale.
- Designed, implemented, and maintained the team's ETL Spark jobs that ran on over a billion rows daily.
- Used technologies, such as Java, Python, Scala, Pivotal Cloud Foundry, Spark, and Hadoop, to build the firm's Insights system.
- Built production debugging dashboards using Dynatrace and Splunk that reduced the time to identify and resolve incidents.
- Served as a technical interviewer for software engineering candidates across the firm.
- Gathered requirements and prioritized features to meet clients' and stakeholders' expectations.

Senior Associate – Software Engineer

02/2020 – 02/2021

JPMorgan Chase & Co.

- Scaled the team's application to over 55 million customers while maintaining an SLO of 250 milliseconds as we surpassed more than 425 transactions per second.
- Selected for the firm's initiative to automate the processing of the SBA PPP Loans, helping distribute \$30 billion to small businesses during the COVID-19 pandemic.
- Led the development of multiple Scala-based Spark jobs that performed analytics and generated KPI reports from our primary application.
- Selected from approximately 30 participants for a weeklong professional program to better understand how to grow as a leader.

Associate – Software Engineer

10/2017 – 02/2020

JPMorgan Chase & Co.

- Built a Java Spring Boot application from scratch that aggregated data from multiple Cassandra data stores to give customers financial insight into their spending and investing habits.
- Scaled the application from supporting hundreds of customers to supporting millions by migrating it to the cloud.
- Created the proof of concept that was used as the basis for moving the application into a cloud environment.
- Helped create a CI/CD pipeline for the application, which reduced our deployment time to production from a few hours to approximately 30 minutes.
- Created the first CI/CD pipeline for Spark jobs within our division at the firm; this became the standard for all CI/CD Spark job pipelines within our division.

PROJECTS

Personal AI Assistant <i>An Agentic AI Project</i> - Built a multi-agent AI assistant using the supervisor/sub-agent architecture, orchestrating 7 specialized agents with local LLMs that have the following skills: task management, exercise tracking, weight tracking, weather, email, date and timezone computations, and STEM computation. - Designed a fully local inference pipeline running on constrained hardware (2GB VRAM), using sliding-window context management, agent-as-tool composition, and a YAML document store.	Python, LangChain, Ollama
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TECHNOLOGIES & SKILLS

Languages: Java, Python, Racket, Scala
Batch Processing & Streaming: AWS EMR, Hadoop, Kafka, PubSub, Spark
Protocols & Frameworks: gRPC, Protobufs, Spring Boot, LangChain
Databases: Cassandra, DuckDB, SQL
Cloud: AWS, Pivotal Cloud Foundry
Other Tools: Git, Linux
Specialties: Big Data Processing, Functional Programming, Highly Distributed Systems, Microservices, NoSQL, Object-Oriented Programming, RESTful Services

LANGUAGES

Chinese (Mandarin): Limited Working Proficiency
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EDUCATION

Illinois Institute of Technology <i>Master of Science and Bachelor of Science in Computer Science; Minor in Applied Mathematics</i>	<i>August 2017</i>
Cornell University <i>Machine Learning Certificate</i>	<i>September 2019</i>
East China Normal University <i>Accelerated Chinese Language Program</i>	<i>Summer 2014</i>